Smart EV Battery Management System

Sample Telemetry Data & Verification Test Cases

Version 1.0 | Synthetic engineering test package | 27 October 2026

Scope: LiFePO4 pack; voltage/current/temperature acquisition; SoC, SoH, SoP; active balancing; protection/isolation; CAN telemetry; fault-state transitions.

Important: All values and thresholds in this document are **illustrative test fixtures, not approved safety limits**. Calibrate to the selected cells, AFE, contactors/MOSFETs, pack design and hazard analysis. No tests here establish automotive safety certification or guarantee thermal-runaway prevention.

1. Test Bench & Assumptions

Parameter	Illustrative fixture / setup
Cell chemistry	LiFePO4, 16 cells in series (16S1P)
Nominal cell / pack	3.2 V / 51.2 V; nominal capacity 100 Ah
Cell operating voltage test band	2.50-3.65 V per cell (example only)
Warning / trip examples	High cell: warn 3.55 V, trip 3.65 V; low cell: warn 2.70 V, trip 2.50 V
Temperature examples	Warn at 50 C; critical at 60 C; rate-of-rise trip above 8 C/min
Current examples	Charge limit +100 A; discharge limit -150 A; convention: positive = charging
Acquisition / telemetry	Cell acquisition target 10 ms; CAN publishing 100 ms
Balancing entry / exit	Start when max-min cell voltage exceeds 30 mV; stop below 10 mV
Equipment	Programmable cell simulator, calibrated DMM, current source/load, thermal simulator, oscilloscope, CAN logger, isolated bench supply

Use low-energy simulation for fault injection first. High-current switching, short-circuit and thermal-abuse tests require approved lab controls and an independent safety plan.

2. Synthetic Sample Data

Values are illustrative snapshots; Vmin/Vmax are cell voltages. SoC is an algorithm output, not independently measured ground truth.

ID	Mode	Vmin / Vmax (V)	I (A)	Tmax (C)	SoC (%)	Expected behavior
D01	Idle	3.298 / 3.306	0	28	75	NORMAL; no balancing
D02	Charge	3.420 / 3.462	+40	33	83	NORMAL; balancing eligible
D03	Discharge	3.120 / 3.140	-80	39	55	NORMAL
D04	High V warning	3.460 / 3.570	+25	36	94	WARNING; limit charge
D05	High V trip	3.470 / 3.670	+20	37	96	FAULT; open charge path
D06	Low V warning	2.680 / 2.740	-60	42	12	WARNING; derate discharge
D07	Low V trip	2.480 / 2.630	-40	44	5	FAULT; open discharge path
D08	Hot module	3.250 / 3.280	-30	61	62	FAULT; isolate per safety design

ID	Mode	Vmin / Vmax (V)	I (A)	Tmax (C)	SoC (%)	Expected behavior
D09	Rapid heating	3.260 / 3.300	-25	48	66	FAULT if dT/dt > 8 C/min
D10	AFE lost	N/A	0	N/A	N/A	FAULT; sensor communication loss

3. Functional Test Case Matrix

ID	Test	Stimulus / steps	Expected result
TC-01	Startup/self-test	Power-cycle MCU with healthy simulated cells	INIT -> NORMAL; self-test and diagnostics logged
TC-02	Cell voltage accuracy	Sweep each input 2.5-3.65 V; compare DMM	Error within validated hardware accuracy budget
TC-03	Acquisition cadence	Timestamp 1000 consecutive cell scan cycles	Target <=10 ms between complete scans, no missed samples
TC-04	Current measurement	Inject -150, -10, 0, +10, +100 A equivalent	Correct polarity, linearity and calibrated tolerance
TC-05	Temperature monitoring	Sweep NTC simulator from 20 to 65 C	Monotonic reading; warning/trip at configured thresholds
TC-06	SoC integration	Apply known Ah charge/discharge profile	SoC moves with correct sign; bounded 0-100%; compare reference
TC-07	SoC rest correction	Simulate rest and chemistry-specific OCV input	Filter converges without discontinuous unsafe SoC jump
TC-08	SoH calculation	Replay capacity/resistance aging datasets	SoH trends correctly; no invalid >100% or negative outputs
TC-09	SoP derating	Raise temperature and reduce minimum voltage	Charge/discharge power limit decreases as configured
TC-10	Balancing start	Set cell spread 42 mV in permitted mode	Balancer starts on eligible high cell(s)
TC-11	Balancing stop	Reduce spread below 10 mV	Balancer stops; no oscillatory rapid switching
TC-12	Balancing inhibit	Request balance during critical fault	All balancing outputs disabled
TC-13	High-voltage warning	Ramp one cell above 3.55 V	WARNING; charge limit issued; event recorded
TC-14	High-voltage trip	Ramp one cell above 3.65 V	FAULT; charging disabled and fault latched
TC-15	Low-voltage trip	Ramp one cell below 2.50 V	FAULT; discharge disabled and fault latched
TC-16	Over-temperature trip	Simulate module sensor at 61 C	FAULT; isolation command and VCU alert
TC-17	Rate-of-rise trip	Apply synthetic 9 C/min heating trace	Trip after specified filter/debounce logic
TC-18	AFE communication loss	Disconnect simulated isolated bus	Watchdog timeout; safe state and diagnostic code
TC-19	CAN status & faults	Capture normal, warning and fault frames	Signal encoding, CRC/counter (if used), period verified
TC-20	Fault recovery	Clear input fault then request reset	Return to NORMAL only after explicit safe recovery criteria
TC-21	MCU reset/watchdog	Freeze task or force reset in bench simulation	Outputs fail safe; restart diagnostic retained
TC-22	Sensor stuck / implausible	Hold NTC constant during changing references	Plausibility diagnostic triggered if supported
TC-23	Power interruption	Remove and restore auxiliary supply	Safe output defaults; nonvolatile events preserved
TC-24	Contactor feedback mismatch	Command off, simulate feedback still closed	Welded/stuck device diagnostic; inhibit restart

4. Timing & Quantitative Acceptance Worksheet

Metric	Example acceptance target	How to verify
Full cell scan period	<=10 ms under maximum system load	Oscilloscope GPIO timestamp + trace log
Protection decision latency	Bounded by safety requirements; record measured worst case	Stimulus edge to decision GPIO
Isolation switching latency	<1 ms only if validated for actual device and current	Command to measured current interruption
Balancing efficiency	>88% at defined voltage/current operating points	Calibrated energy in/out over test interval
CAN status period	100 ms +/- configured jitter tolerance	CAN analyzer timestamps
SoC estimation error	Set target from drive cycle and reference instrumentation	Compare integrated Ah / reference SoC

Note: The document specification mentions <1 ms switching and +/-1 mV sensing. Those are design targets that must be demonstrated across temperature, load, noise and fault conditions; 12-bit ADC resolution alone does not establish +/-1 mV accuracy.

5. Detailed Example: TC-14 Overvoltage Trip

Field	Example
Preconditions	AFE healthy; charger emulator enabled; all cells 3.40 V; controller NORMAL
Steps	1. Raise cell 7 to 3.56 V: observe warning. 2. Raise to 3.67 V: observe trip. 3. Lower to 3.40 V without reset. 4. Request authorized recovery.
Expected	Charge path disabled at trip; timestamp and cell index logged; CAN fault sent; no automatic unsafe reclose; recovery only after configured criteria.
Record	Trip threshold, debounce time, output-command time, measured current interruption, fault code, reset result.
Pass rule	All safety actions meet approved calibrated requirements; no unintended charging after trip.

6. Example Execution Log (synthetic, not measured)

Time (ms)	Cell 7 (V)	State	Charge allowed	Fault
0	3.400	NORMAL	Yes	None
100	3.560	WARNING	Derated	OV_WARN
200	3.670	FAULT	No	OV_TRIP
250	3.400	FAULT (latched)	No	OV_TRIP
500	3.400	NORMAL after valid reset	Yes	Cleared / archived

7. Test Record Template

Field	Fill during execution
Test case / revision	_____
Date / operator / bench ID	_____

Field	Fill during execution
Firmware / hardware revision	_____
Instrument calibration IDs	_____
Input conditions / steps	_____
Expected vs. actual results	_____
Timing / traces / log references	_____
Pass / Fail / Blocked	_____
Defect ID / reviewer signoff	_____

8. Safety & Coverage Notes

Include boundary-value testing, sensor open/short, supply brownout, stuck-on balance switch, isolation device failure, simultaneous faults, EMI robustness, thermal gradients, current-sensor drift, CAN timeout, corrupted configuration and safe firmware update. Separate software-in-the-loop, hardware-in-the-loop and controlled physical-pack validation.

A battery disconnect cannot by itself guarantee prevention of internal cell thermal runaway or propagation. Design must additionally address passive propagation barriers, fusing, venting, thermal sensing coverage and vehicle-level emergency behavior. Map each requirement to test IDs, evidence and a hazard analysis (e.g., ISO 26262 and relevant battery standards as applicable).